

Dhruv Patel

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Technical Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL, MATLAB, R
AI / ML: PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, LangChain, LangGraph, HuggingFace, FAISS, CUDA, OpenRouter, MCP
Backend / Web: Next.js, React, FastAPI, Flask, Node.js, REST APIs, WebSockets, Server-Sent Events
Systems: Distributed Systems, Concurrency, Multithreading, Socket Programming, System Design, Object-Oriented Design, Performance Optimization, Unix/Linux
Cloud / DevOps: Docker, Kubernetes, AWS, Azure, Google Cloud Platform, Git, GitHub, CI/CD, pytest, JUnit, Vercel
Databases: PostgreSQL, Firestore, MongoDB, Redis, MySQL, Supabase

Experience

- **Software Engineer** Jun 2026 – Present
Aaron Technologies Inc. — Sole Engineer, B2B Industrial Sourcing Platform Remote
 - **Architected and shipped a serverless backend** using Firestore, Resend, and HubSpot CRM through Next.js API routes to handle data persistence, transactional email, and CRM lead synchronization for a production B2B platform.
 - Built an interactive **landed-cost calculator** modeling freight, duty, and handling, alongside 15+ responsive pages across product, catalog, industry, and RFQ experiences using Next.js, React, and TypeScript — shipped using **AI-assisted development** workflows.
 - Remediated site-wide **accessibility (WCAG 2.1 AA)**, security, SEO, and performance, including contrast and heading-hierarchy fixes, JSON-LD structured data, and production deployment hardening.
- **Research Project** Jan 2024 – Jul 2024
Accelerating Particle-in-Cell Simulations Using Deep Learning (APS GEC 2025) DA-IICT, India
 - Accelerated device-scale **Particle-in-Cell (PIC-MCC) plasma simulations** by replacing the charge-deposition, Poisson-solver, and electric-field stages — the dominant parallel bottleneck — with a **U-Net surrogate model, cutting execution time 45%** on 128x128 grids with 2.6e5 particles.
 - Enforced physical fidelity using **Physics-Informed Neural Networks**, holding mean percentage error **under 5%** against baseline PIC-MCC across 500-iteration ExB plasma test cases; implemented in Python with C extensions on Unix/Linux, optimizing memory allocation and numerical computation.
- **Summer Research Intern** May 2023 – Aug 2023
Software Testing and Test Case Prioritization DA-IICT, India
 - Implemented and benchmarked four data structure and algorithm-based test-case prioritization techniques (FAST, Hypervolume-based Genetic, Bat-inspired, Lexicographical Ordering) across **30 open-source Java projects** ranging up to 87K SLOC and 5,000+ test cases, using runtime and mutation-based APFD.
 - Profiled execution bottlenecks and automated evaluation pipelines in Python, **reducing evaluation pipeline runtime by 30%**.

Projects

- **Raft Consensus & Multi-Paxos Distributed Transaction Sequencer** C++, Distributed Systems
– Implemented **Raft consensus** (leader election, replicated logs, fault-tolerant recovery) for a distributed key-value system, validated under simulated node failures and network partitions.
– Extended the system with a deterministic-ordering transaction engine using a global sequencer and **Multi-Paxos consensus** to guarantee strictly serializable execution; optimized the scheduler's hot path to sustain **200+ TPS under 3-replica consensus** (390+ TPS single-replica baseline).
- **ConsensusPrompt: Multi-Agent Middleware for Prompt Optimization** LangGraph, FastAPI, Next.js, OpenRouter
– Engineered a multi-agent middleware applying anonymous peer review to prompt engineering, coordinating rewriter, reviewer, and chairman agents via LangGraph.
– Validated the system in a **13-user study** achieving 4.77/5 mean trust with a 0% override rate; an ablation study showed a **25.5% relative gain** in consensus strength from a 5-model reviewer configuration.
- **ARGUS: Local AI Site Compliance Platform** Python, FastAPI, CUDA, PyTorch, React
– Architected a fully on-device, privacy-preserving **multimodal AI platform** orchestrating five models (Cosmos-Reason2-8B VLM, CLIP ViT-L/14, Faster R-CNN ResNet50-FPN, Qwen3 embedding, Nemotron reranking) to ground construction-site video findings in OSHA and NYC DOB regulations via retrieval-augmented search.
– Integrated **7 live NYC Open Data datasets** (permits, violations, complaints, incidents) for compliance enrichment; optimized local **CUDA** inference via in-process detector reuse, VLM batching, SDPA attention, and startup prewarming.
- **OnboardOps: AI-Powered Repository Onboarding Platform** FastAPI, Next.js, MCP, GitHub API
– Built an AI-powered developer onboarding platform with a **custom MCP server** and FastAPI backend to automate understanding of unfamiliar, large software systems.
– Engineered real-time WebSocket workflows and repository information retrieval using GitPython and GitHub APIs to identify dependency graphs, entry points, code hotspots, and project conventions.
- **Freight Room: Multi-Agent Supply Chain Response Platform** Winner – Band of Agents Hackathon 2026
– Built a distributed platform that detects vessel delays from a **live AIS feed** and dynamically recruits **7 specialized agents across 3 frameworks** (LangGraph, PydanticAI, CrewAI) via Band to coordinate logistics incident response.
– Designed an auditable human-in-the-loop workflow — quorum voting, adversarial dissent review, and demurrage estimation grounded in published carrier tariffs — emitting traceable decision dossiers from full room history.

Education

- **Stony Brook University** Aug 2025 – May 2027
Master of Science in Computer Science, GPA: 3.61/4.00 Stony Brook, NY
- **Dhirubhai Ambani University (Formerly DA-IICT)** Nov 2020 – May 2024
B.Tech. (Honours) in ICT, Minor in Computational Science, GPA: 3.45/4.00 Gandhinagar, India

Leadership

- **Deputy Convenor** Apr 2021 – Apr 2023
Annual Techno-Cultural Festival Committee DA-IICT, India
 - Led the technical team for one of Gujarat's largest techno-cultural festivals, delivering infrastructure and event execution for 25,000+ attendees.